INCH-POUND

MS25270L
26 June 2012
SUPERSEDING
MS25270K
27 November 2003

DETAIL SPECIFICATION SHEET

# RELAYS, ELECTROMAGNETIC, 5 AMPERES, 6 PDT, TYPE I, POTTED LEAD, HERMETICALLY SEALED

INACTIVE FOR NEW DESIGN AFTER 5 JUN 1987. NO SUPERSEDING SPECIFICATION.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

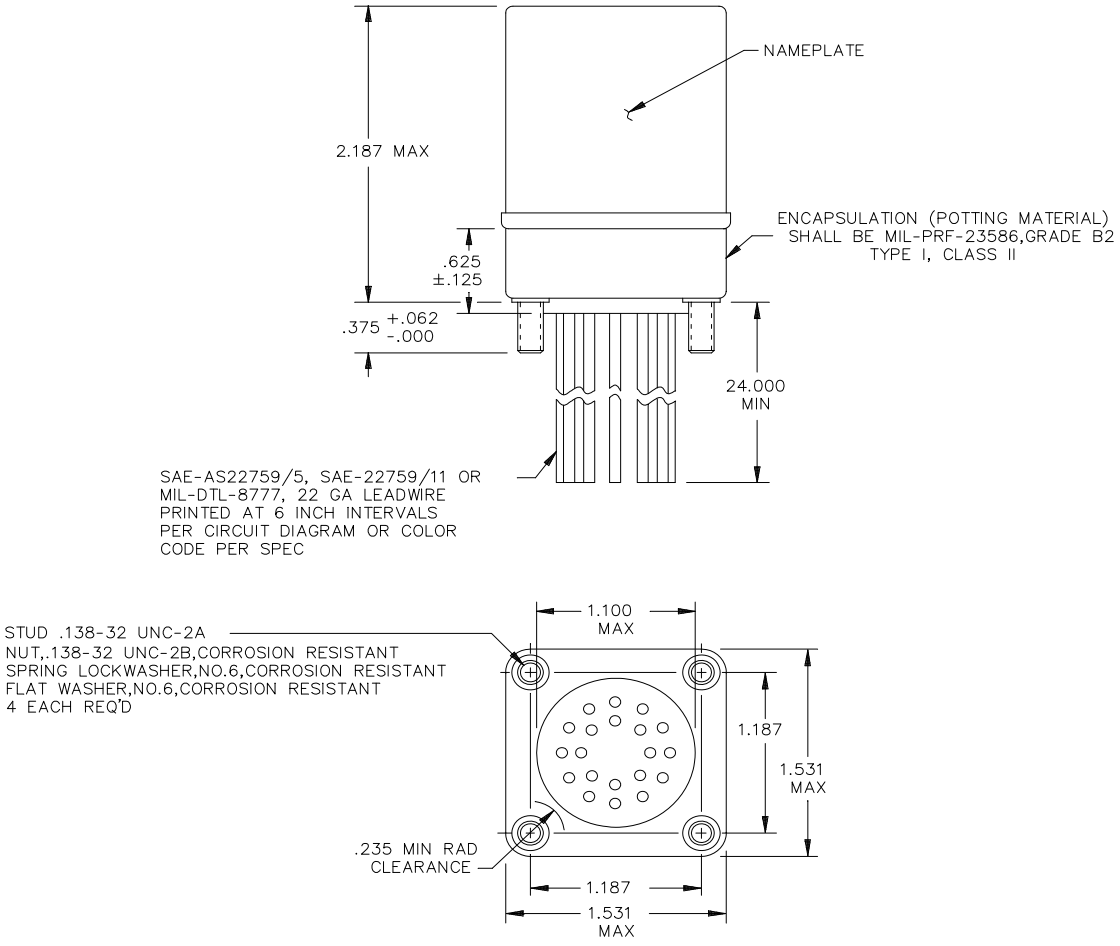
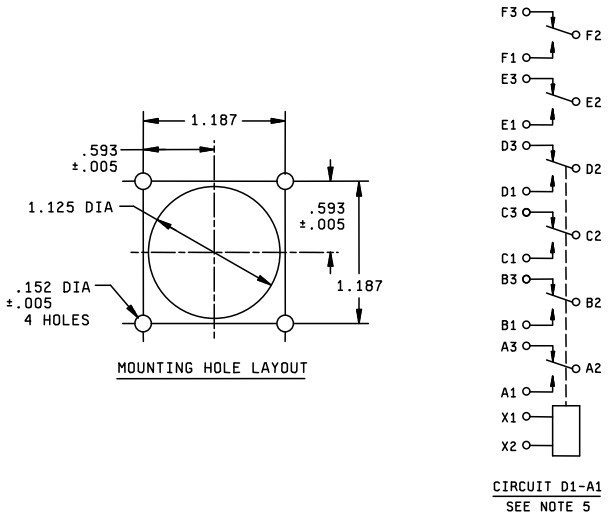



FIGURE 1. Dimensions and configurations.



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NOTES:

1/ Dimensions are in inches.

2/ Metric equivalents are given for general information only.

3/ Unless otherwise specified, tolerance is ±.010 (0.25 mm).

4/ Terminal numbers need not appear on relay header provided there is affixed to the relay a suitable legible circuit diagram that permanently and positively identifies each terminal location specified hereon.

5/ The use of diodes on ac relays is optional. Actual application must be shown on label.

6/ In the event of conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

| Inches | mm |
|---|---|
| .005 | 0.13 |
| .062 | 1.57 |
| .125 | 3.18 |
| .152 | 3.86 |
| .235 | 5.97 |
| .375 | 9.53 |
| .025 | 15.88 |
| 1.100 | 27.94 |
| 1.187 | 30.15 |
| 1.531 | 38.89 |
| 2.187 | 55.55 |
| 24.00 | 609.60 |

FIGURE 1. Dimensions and configurations – Continued.

TABLE I. Dash numbers and characteristics.

| Dash number MS25270- | Type | Coil | Terminal type | Mounting or mating socket | Max weight in pounds |
|---|---|---|---|---|---|
| D1 | I | dc | Lead | Stud | .67 |
| A1 | I | ac | Lead | Stud | .67 |



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TABLE II. Operating characteristics.

| PIN MS25270- | Coil data | | | | | | | | | | | Time - milliseconds max | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Coil | Rated | | | Max | | 1/ Max pick-up voltage | | | Hold voltage 2/ | Drop out voltage 2/ | Operate 3/ | Release 4/ | Contact Bounce | | | |
| | | | | | | | | | | | | | | Main | | Aux | |
| | | Volts 1/ | Freq Hz | Ω Res | Volts | Amp | Normal 2/ | High temp test | Cont current test | | | | | NO | NC | NO | NC |
| D1 | X1, X2 | 28 | dc | N/A | 29 | 0.18 | 18 | 19.8 | 22.5 | 7.0 | 1.5 | 20 | 20 | 2 | 2 | N/A | N/A |
| A1 | X1, X2 | 115 | 400 5/ | N/A | 122 | 0.04 | 90 | 95 | 103 | 30 | 5.0 | 25 | 50 | 2 | 2 | N/A | N/A |

1/ CAUTION: Use of any coil voltage less than rated coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.
5/ MS25270-A1 may be used on 60 Hz if maximum ambient temperature is +85°C (maximum coil current shall be 0.044 ampere).

TABLE III. Rated contact load (amperes per pole) (case grounded).

| Type of load | Life operating cycles x $10^3$ | 28 V dc | | | | 115 V ac, 1 phase | | | | 115/200 V ac, 3 phase 1/ | | | | See appropriate notes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main | | Aux | | |
| | | NO | NC | NO | NC | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | |
| Resistive | 100 | 5 | 5 | | | 5 | 4 | | | | | | | |
| Inductive | 100 | | | | | | | | | | | | | |
| Inductive | 20 | 3 | 3 | | | 3 | 2 | | | | | | | |
| Motor | 100 | 1.5 | 1.5 | | | 1.5 | 1 | | | | | | | |
| Lamp | 100 | 0.8 | 0.8 | | | 0.8 | 0.6 | | | | | | | |
| Transfer load | | | | | | | | | | | | | | 2/ |
| Mechanical life reduced current | 400 | 1.25 | 1.25 | | | 1.25 | 1 | | | | | | | |
| Mixed loads | Applicable per specification | | | | | | | | | | | | | |

1/ Absence of value indicates relay is not rated for 3-phase application.
2/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.

Environmental characteristics.

| | |
|---|---|
| Temperature range | -70°C to +125°C |
| Max altitude rating | 80,000 ft |
| Shock G-level | 25 g's |
| Duration | 11 ms |
| Max duration contact opening | 10 μs |
| Vibration - sinusoidal (see chart below) | |
| G-level | 10 g's |
| Frequency range | 5-1,500 Hz |



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| | |
|---|---|
| Vibration - random | |
| Applicable spec | N/A |
| Power spectral density | N/A |
| RMS G min | N/A |
| Frequency range | N/A |
| Curve | N/A |
| High shock | N/A |
| Acceleration | 15 g's |

Electrical characteristics.

| | |
|---|---|
| Minimum insulation resistance, initial | 100 megohms. |
| After life or environmental tests | 50 megohms. |

Dielectric strength (sea level).

| | Initial | After life tests |
|---|---|---|
| Coil to case | 1,050 V rms | 1,000 V rms |
| Aux contacts | | |
| All other points | 1,050 V rms | 1,000 V rms |

Dielectric strength (altitude).

| | 80,000 ft |
|---|---|
| Coil to case | 1,000 V rms |
| Aux contacts | |
| All other points | 1,000 V rms |

| | |
|---|---|
| Max contact drop initial | 0.150 volt. |
| After life test | 0.175 volt. |
| Overload current | 20 amperes. |
| Rupture current | 25 amperes. |
| Duty rating | Continuous. |
| RFI specification | MIL-STD-461 |

(Applicable to coil circuits of ac operated relays).

Conformance inspection.

Performance of groups B and C tests are not applicable.

Group A acceptance reports shall be submitted to the preparing activity on a yearly basis in order to retain qualification for this military standard sheet.

Qualification by similarity: See MIL-PRF-6106.



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NOTES

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-6106, this document references the following:

MIL-DTL-8777 MIL-PRF-23586 SAE-AS22759/5 SAE-AS22759/11 MIL-STD-461

Custodians:
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2012-009)

Review activities:
Navy - EC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



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